

ASHASHREE SARMA

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EDUCATION

International Institute of Information Technology, Bangalore <i>M.S. by Research in Data Science CGPA: 3.55/4.0</i>	Aug 2024 – Present <i>Bangalore, India</i>
Assam Engineering College, Assam Science and Technology University <i>B.E. (Honors) in Computer Science & Engineering First Class, 81.4% (Top 5%)</i>	Aug 2019 – June 2023 <i>Guwahati, India</i>
Maharishi Vidya Mandir Public School <i>Class XII (CBSE) 87.6% (Grade: A)</i>	June 2017 – May 2019 <i>Guwahati, India</i>
SBOA Public School <i>Class X (CBSE) 10.0 CGPA</i>	March 2016 – Feb 2017 <i>Guwahati, India</i>

EXPERIENCE

Indian Institute of Technology Guwahati (IIT Guwahati) <i>Deep Learning Research Intern, Da'Spatio Rhobotique Lab, Technology Incubation Centre</i>	Feb 2024 – July 2024
- Implemented and benchmarked CNN-based encoder-decoder and Vision-Language Model (VLM) architectures for an image captioning task, evaluating model outputs using standard metrics (BLEU, CIDEr) under the guidance of Scholar Deeksha Varshney. - Fine-tuned the GIT (Generative Image-to-Text) transformer model on domain-specific image-caption datasets using Hugging Face Transformers, systematically comparing performance across pre-trained model variants to identify optimal architectures. - Preprocessed and managed multimodal datasets (image-caption pairs) using Hugging Face Datasets, ensuring data quality and pipeline compatibility for GPU-accelerated training on institutional compute infrastructure. Tech Stack: Python, PyTorch, Hugging Face Transformers & Datasets, CUDA, GPU Acceleration.	
Nestria (Byte) – Spryte Labs <i>Data Science & Software Engineering Intern</i>	Aug 2023 – Dec 2023 <i>Remote</i>
- Developed end-to-end machine learning pipelines encompassing data ingestion, preprocessing, feature engineering, model training, and evaluation on real-world business datasets using Scikit-learn and Pandas. - Conducted exploratory data analysis (EDA) on large, multi-dimensional datasets — profiling distributions, detecting outliers, and uncovering feature correlations — and translated findings into data-driven recommendations for stakeholders. - Automated recurring data processing workflows using Apache Airflow DAGs, reducing manual intervention and improving pipeline reliability across scheduled data tasks. Tech Stack: Python, NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, Apache Airflow	

PUBLICATIONS

Modeling Outcomes-led Learner Behavior and Emergent Social Synchrony	2025
Authors: Ashashree Sarma, Sushree Behera, Srinath Srinivasa, Prasad Ram. - Published in <i>Proceedings of the 12th ACM Conference on Learning @ Scale (L@S '25)</i>, Pages 371–375. DOI: 10.1145/3698205.3733959 Venue: ACM Learning @ Scale 2025	

TECHNICAL SKILLS

Programming Languages & Frameworks: Python, JavaScript, HTML/CSS, SQL, PyTorch, Flask, FastAPI
Machine Learning & Data Libraries: NumPy, Pandas, Matplotlib, Scikit-learn, NLTK, Plotly, Seaborn, OpenCV
Developer Tools & Platforms: Git, GitHub, VS Code, MySQL, MongoDB, Apache Airflow, MLOps, CUDA

PROJECTS

Government Policy Education AI Chatbot | *RAG, LLM, Claude 3 Opus, FastAPI, Next.js* March 2024

- Built a full-stack AI chatbot to improve citizen access to government policies using a Retrieval-Augmented Generation (RAG) pipeline with Claude 3 Opus, enabling context-grounded responses to queries on complex financial and civic policy documents.
- Designed the retrieval layer to chunk, index, and query policy documents, injecting retrieved context into LLM prompts to reduce hallucinations and improve factual accuracy of generated responses.
- **Live Demo:** [Render + Vercel Deployment](#) **GitHub:** [Permissionless-AI-Chatbot](#)

Educational Course Recommendation System | *Content-Based Filtering, NLP, Scikit-learn, FastAPI, Next.js* January 2024

- Built a content-based filtering recommendation engine that suggests similar courses given a user-selected course, using TF-IDF vectorization and Cosine Similarity over course metadata to compute pairwise relevance scores.
- Applied NLP preprocessing steps including tokenization, stopword removal, and stemming to normalize course descriptions before vectorization, improving recommendation quality.
- **Live Demo:** [Render + Vercel Deployment](#) **GitHub:** [Educational-Course-RecommendationSys](#)

Contextual Sentiment Analysis | *VADER, NER, NLTK, BeautifulSoup, newspaper3k, Flask* January 2023

- Developed a Flask application that retrieves news articles for a queried entity via the Google Custom Search API and classifies them by sentiment using the VADER lexicon-based NLP model, returning polarity scores (positive, negative, neutral, compound).
- Incorporated POS tagging and Named Entity Recognition (NER) using NLTK to accurately identify and isolate the queried entity within article text, improving sentiment attribution precision at the entity level.
- **GitHub:** [Ethos-Hackathon-Final](#)

CERTIFICATIONS

- Google Data Analytics Professional Certificate | Grade: 92.50% ⇨ Coursera, Google
- Data Science Cohort – Bootcamp ⇨ Byte Academy
- Python for Data Science | Consolidated Score: 72% ⇨ NPTEL, IIT Madras
- Explore ML ⇨ Google AI

LEADERSHIP & ACHIEVEMENTS

- **AI Mayhem Hackathon – Permissionless:** Competed in an AI-focused hackathon, presenting a working AI solution and demonstrating expertise in applied machine learning and rapid prototyping under time constraints. ⇨ March 2024
- **ServiceNow Women Code to Win Hackathon:** Competed in ServiceNow's online coding hackathon, demonstrating proficiency in algorithmic problem-solving and software development. February 2023
- **Ethos Hackathon – IIT Guwahati (Finals):** Led end-to-end development of a news sentiment analysis and product review platform at an All-India Level hackathon, advancing to the national finals held at IIT Guwahati. January 2023
- **Smart India Hackathon 2022 – National Selection:** Led the design and implementation of a secure two-factor authentication (2FA) system as part of the SIH 2022 national-level competition, selected to represent the institution. March 2022
- **AEC Coding Club – Core Team Member:** Drove technical community building as a founding core team member, organizing workshops, coding seminars, and events for 100+ students; maintained club continuity through the COVID-19 pandemic. Sept 2019 – May 2023

EXTRACURRICULAR ACTIVITIES

- **Campus Barta – College Community Progressive Web App:** Led the full-stack development of a PWA for the college campus, building a community announcement board, and integrating a live news feed via the News API. ⇨ May 2022
- **Technical Outreach Program for Government Schools:** Designed and delivered a weekly technology education program teaching internet literacy and web development fundamentals to primary school students, improving access to technology education in underserved communities. May – July 2022

- **Coordinator – Pyrokinesis (Annual College Fest):** Managed end-to-end planning and execution of the college's annual festival, coordinating a cross-functional team and ensuring seamless event delivery. **March 2023**
- **Vice President – Mukhaa (Drama & Film Society):** Directed screenplay development, led a creative team in producing zero-budget short films and street performances, fostering artistic expression and team collaboration. ➡ **June 2021 – July 2022**